Blockchain Enabled Secure Medical Delivery Robot Using IoT

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***Abstract* — Medicine delivery in hospitals is a routine yet critical task that requires accuracy, timely distribution, and secure handling of medication. Traditional medicine delivery systems rely heavily on manual intervention by healthcare staff, which can lead to increased workload, delays in delivery, and potential human errors in medication management. In this paper, we propose a Blockchain Enabled Secure Medical Delivery Robot Using IoT designed to automate the medicine delivery process within hospital environments. The proposed system uses an ESP8266 NodeMCU microcontroller to control a robotic platform that navigates along a predefined path to deliver medicines to designated patient locations. A secure RFID-based authentication mechanism ensures that only authorized individuals can access the medicine compartment. Once authenticated, a servo motor-controlled mechanism opens the compartment, while an infrared sensor detects whether the medicine has been successfully retrieved. The system transmits delivery status and health monitoring data to a cloud-based IoT platform for real-time monitoring through a web dashboard. To enhance data integrity and prevent unauthorized modification of medical records, a blockchain-inspired SHA-256 hashing mechanism is implemented. Experimental results demonstrate reliable robot navigation, secure authentication, and efficient IoT-based monitoring. The proposed architecture provides a secure, contactless, and automated medicine delivery solution, improving hospital efficiency, reducing manual workload, and enhancing patient safety in modern smart healthcare environments.**

Keywords — Internet of Things (IoT), Medical Delivery Robot, RFID Authentication, Blockchain Security, ESP8266 NodeMCU, Smart Healthcare, SHA-256 Hashing, Healthcare Automation.

I. Introduction

Medicine delivery is one of the most common and essential tasks in hospital environments for patient treatment and healthcare management. Ensuring that patients receive the correct medication at the right time requires continuous monitoring and coordination by healthcare staff. Traditionally, medicines are delivered manually by nurses or hospital assistants who collect medications from the pharmacy and distribute them to patients across different wards. Although this approach has been widely practiced, it increases the workload of healthcare professionals and relies heavily on human supervision. In busy hospital environments, particularly in intensive care units (ICUs) and high patient-density wards, maintaining timely and accurate medicine delivery can be challenging. Delays in delivering medication or errors in distribution may lead to serious medical complications and affect the quality of patient care.

With the growing demand for efficient healthcare services, hospitals require intelligent systems that can automate routine tasks and improve operational efficiency. Automated medicine delivery systems using robotics and Internet of Things (IoT) technologies can assist healthcare staff by reducing manual workload and ensuring timely distribution of medication. In addition, secure authentication mechanisms and reliable monitoring systems are required to prevent unauthorized access to medicines and maintain accurate medical records. Therefore, the development of an intelligent and secure medicine delivery system has become an important requirement for modern smart hospital environments.

II. Related Work

A. IoT-Based Medical Delivery Systems

Several studies have explored the use of Internet of Things (IoT) technologies for automating hospital logistics and medicine distribution. In many IoT-based healthcare automation systems, microcontrollers and wireless communication modules are used to monitor and control medical equipment remotely. Some robotic delivery systems utilize embedded controllers and wireless communication technologies such as Wi-Fi, Bluetooth, or GSM modules to transport medicines and medical supplies within hospital environments. These systems often include basic navigation mechanisms and sensor-based monitoring to ensure that medicines are delivered to the correct locations.

Although these systems help reduce manual workload and improve operational efficiency, many of them lack secure authentication mechanisms to prevent unauthorized access to medicines. In addition, most existing robotic delivery solutions focus mainly on transportation and do not incorporate advanced data security techniques to protect sensitive healthcare information. The absence of secure access control and reliable data protection mechanisms limits their effectiveness in modern smart hospital infrastructures.

B. RFID-Based Authentication Systems in Healthcare

RFID technology has been widely used in healthcare systems for identification, tracking, and secure access control. Many hospitals use RFID tags to identify patients, track medical equipment, and manage pharmaceutical inventory. In authentication-based systems, RFID readers verify unique identification tags assigned to authorized users before granting access to restricted resources.

While RFID-based systems provide fast and contactless authentication, many implementations are limited to identification purposes only and are not integrated with automated robotic delivery systems. Furthermore, these systems often lack real-time monitoring capabilities and secure data storage mechanisms. Without proper integration with IoT platforms and secure data management technologies, RFID systems alone cannot fully address the challenges associated with automated medicine delivery and healthcare data protection.

C. Blockchain-Based Security in Healthcare Systems

Blockchain technology has gained attention in recent years for its ability to provide secure and tamper-resistant data management. In healthcare applications, blockchain-based systems are used to protect patient records, ensure transparency in medical data transactions, and prevent unauthorized modification of sensitive information. Cryptographic hashing algorithms such as SHA-256 are commonly used to maintain the integrity of stored data.

Despite the advantages of blockchain technology in healthcare security, many existing implementations focus mainly on electronic health records and data storage rather than integrating blockchain mechanisms with real-time healthcare automation systems. The combination of robotics, IoT communication, secure authentication, and blockchain-based data protection is still relatively unexplored in healthcare logistics applications.

III. SYSTEM ARCHITECTURE AND METHODOLOY

A. Overall System Architecture

The proposed project titled **Blockchain Enabled Secure Medical Delivery Robot Using IoT** is designed using an integrated architecture that combines **robotic automation, IoT communication, authentication mechanisms, and blockchain-inspired data security**. The system consists of a **robotic delivery unit**, an **authentication and sensing module**, and a **cloud-based monitoring platform**. This architecture enables secure medicine delivery while allowing healthcare professionals to monitor delivery status and patient health parameters remotely.

The **ESP8266 NodeMCU microcontroller** serves as the central processing and communication unit of the system. It controls the movement of the robotic platform, processes data from sensors, and manages communication with the IoT cloud platform through its built-in Wi-Fi module. The robotic system follows a predefined path to reach the medicine delivery location where authentication and medicine retrieval take place.

The architecture ensures that medicines are delivered efficiently while maintaining **secure access control, real-time monitoring, and reliable data storage**. The integration of IoT communication with blockchain-inspired security mechanisms improves system transparency, reliability, and data integrity within hospital environments.

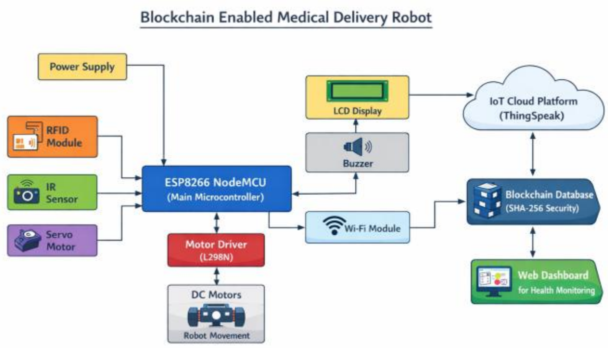

Fig. 1. Block diagram of the Blockchain Enabled Medical Delivery Robot system

B. Robotic Delivery Unit Design and Data Acquisition

The robotic delivery unit is developed around the ESP8266 NodeMCU microcontroller, which interfaces with various hardware components responsible for robot navigation, authentication, and monitoring. The robot uses DC motors connected through an L298N motor driver module to move along a predefined delivery path within the hospital environment.

To ensure secure access to medicines, the system incorporates an **RFID-based authentication mechanism**. Each authorized user is assigned a unique RFID tag that is verified by the RFID reader connected to the NodeMCU. When the tag is scanned, the system checks the identification data before granting access to the medicine compartment.

A **servo motor mechanism** controls the opening and closing of the medicine compartment. After successful authentication, the servo motor rotates to open the compartment door, allowing the patient or authorized person to retrieve the medication. An **infrared (IR) sensor** is installed inside the compartment to detect whether the medicine has been removed.

The NodeMCU continuously collects sensor data and processes authentication results locally. This embedded processing enables quick detection of delivery completion and immediate updating of delivery status.

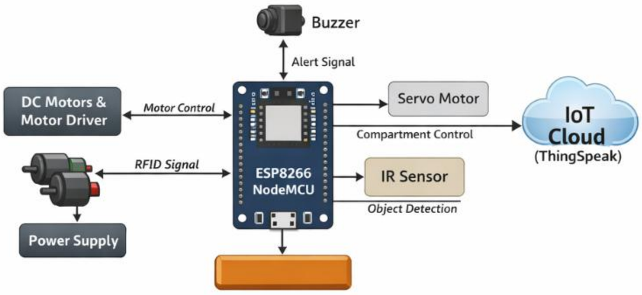

Fig. 2. Functional block diagram of robotic delivery unit

C. Wi-Fi Communication and IoT Network

The communication architecture of the proposed system uses the **built-in Wi-Fi capability of the ESP8266 NodeMCU** to establish a connection with a cloud-based IoT platform such as **ThingSpeak**. Sensor readings, delivery status, and health monitoring data are transmitted to the cloud through HTTP protocols and stored for real-time monitoring and record management.

This IoT-based communication framework allows healthcare professionals to monitor the medicine delivery process remotely through a **web-based dashboard**. The system ensures reliable communication with minimal latency so that hospital staff can respond quickly to delivery notifications or abnormal system events.

Even during temporary network interruptions, the robotic unit continues to operate locally and stores data until the network connection is restored. This ensures that the system maintains reliable operation without affecting the delivery process.

D. Alert Mechanism and Secure Access Control

The proposed system incorporates a **multi-layer alert and notification mechanism** to ensure reliable operation and secure medicine distribution. When the robot reaches the delivery location, the system waits for RFID authentication. If an authorized RFID tag is detected, the medicine compartment is opened automatically.

If the medicine is not retrieved within a predefined time period, the system activates an **audio alert using a buzzer** to notify nearby healthcare staff. Additionally, the delivery status is updated on the IoT platform so that hospital personnel can track whether the medicine has been taken.

To enhance data security, the system uses a **blockchain-inspired SHA-256 hashing mechanism** to secure medicine delivery records and health monitoring data. Any modification to stored data results in a change in the hash value, enabling detection of unauthorized tampering. This approach ensures reliable and secure data management within healthcare systems.

E. Web Server Integration

The proposed system integrates a **web-based monitoring platform** developed using **HTML, CSS, and JavaScript** for user interface design. The platform retrieves data from the **ThingSpeak IoT cloud platform** and displays information such as medicine delivery status and patient health parameters in real time.

Healthcare professionals can access the web dashboard from remote locations to monitor system activity, verify delivery completion, and track patient health data. The platform also stores historical records of delivery events, enabling analysis and documentation of healthcare operations.

The combination of **robotic automation, IoT communication, authentication mechanisms, and blockchain-inspired security** provides a scalable and reliable healthcare automation system suitable for modern smart hospitals.

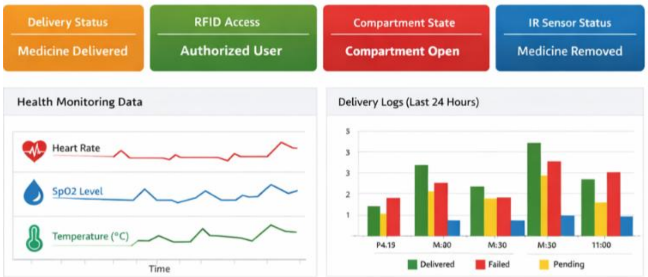

Fig. 3. Web dashboard for IoT-based monitoring system

IV. SOFTWARE DESCRIPTION AND PROGRAM IMPLEMENTATION

The proposed **Blockchain Enabled Secure Medical Delivery Robot Using IoT** is implemented using a combination of embedded programming, IoT communication libraries, and web-based monitoring software. The software framework enables reliable robot navigation control, RFID authentication processing, sensor data acquisition, IoT communication, and secure data management. The system software consists of two main components: the **firmware running on the ESP8266 NodeMCU microcontroller** and the **web-based monitoring application for remote supervision**.

The firmware manages robot movement, authentication verification, sensor monitoring, and communication with the IoT platform. The web-based application receives and visualizes system data such as medicine delivery status and health parameters in real time. Together, these components provide a reliable and intelligent healthcare automation system.

A. *Software Architecture*

The software architecture of the proposed system is designed using a **modular and layered approach**, which improves reliability, scalability, and ease of maintenance. Each functional module is developed independently and integrated into the overall system for coordinated operation.
The main software modules include:

- Robot movement control module
- RFID authentication processing module
- Sensor data acquisition module
- Wi-Fi communication module
- IoT cloud data transmission module
- Data security module using SHA-256 hashing
- Web dashboard monitoring module

The embedded firmware running on the **ESP8266 NodeMCU** performs real-time processing of sensor data, authentication verification, robot navigation control, and cloud communication. The web application software stores and visualizes system data through an interactive monitoring dashboard. This modular architecture ensures **stable real-time operation with minimal latency**, which is essential for healthcare applications.

B. *Firmware Unit of ESP8266 Monitoring Unit*

The firmware for the robotic system is developed using **Embedded C/C++ programming in the Arduino IDE**. The **ESP8266 NodeMCU microcontroller** acts as the central controller that interfaces with all hardware components in the system. The firmware controls the following components:

- **RFID Module** for authentication of authorized users
- **Servo Motor** for opening and closing the medicine compartment
- **IR Sensor** for detecting medicine retrieval
- **Motor Driver (L298N)** for robot movement control
- **LCD Display** for showing system status messages
- **Buzzer** for alert notifications

When the robot reaches the delivery location, the system waits for an RFID authentication request. Once a valid RFID tag is detected, the NodeMCU verifies the authentication data and activates the servo motor to open the medicine compartment. The IR sensor then detects whether the medicine has been removed.

The firmware also manages robot navigation by sending control signals to the motor driver module. The program continuously monitors system status and updates the IoT platform accordingly. This firmware design ensures **stable operation, efficient resource utilization, and reliable system performance**.

C. *Web Application Software and Monitoring Logic*

The system includes a **web-based monitoring dashboard** that allows healthcare professionals to monitor medicine delivery and patient health parameters remotely. The web application is developed using:

- **HTML5** for webpage structure
- **CSS3** for user interface design
- **JavaScript** for dynamic data visualization
- **ThingSpeak IoT platform** for real-time data monitoring

The NodeMCU sends system data to the IoT cloud through Wi-Fi communication. The dashboard displays important information such as:

- Medicine delivery status
- RFID authentication results
- Compartment open/close status
- Health monitoring parameters (heart rate, SpO_2, temperature)

The monitoring system also stores historical delivery records, enabling hospital staff to track previous delivery events and analyze system performance.

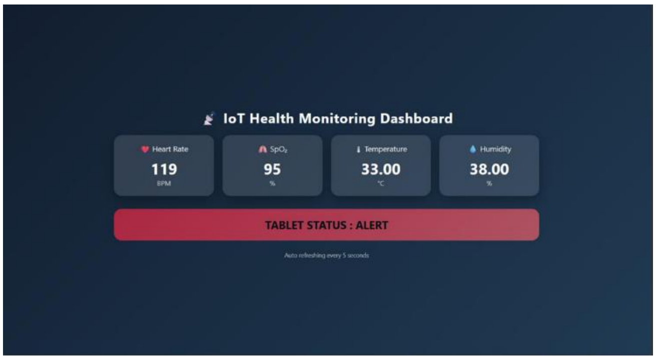

Fig. 4. Web-Based IoT Monitoring Dashboard

D. *Communication Protocol and Data Format*

The proposed system uses a **lightweight HTTP-based communication protocol** to transmit data from the ESP8266 NodeMCU to the IoT cloud platform. The transmitted data includes:

- RFID authentication status
- Medicine delivery confirmation
- IR sensor detection status
- Patient health parameters
- Timestamp and device identification number

The data is formatted into structured packets before being transmitted to the cloud server. This structured communication format ensures that data is processed efficiently by the monitoring platform. Wi-Fi communication parameters are optimized to reduce latency and ensure reliable connectivity within the hospital network.

Event-based data transmission is used for emergency alerts, while periodic updates are sent at regular intervals for continuous monitoring.

E. Algorithm and Flowchart Description

The overall operational algorithm of the proposed system consists of several steps including authentication verification, robot navigation, medicine delivery monitoring, IoT data transmission, and security verification.

First, the ESP8266 NodeMCU initializes all connected sensors and hardware components. The robot then moves along a predefined path to reach the medicine delivery location. Once the robot arrives, the system activates the RFID module to verify the identity of the user.

If the RFID authentication is successful, the servo motor opens the medicine compartment and allows the medicine to be retrieved. The IR sensor detects whether the medicine has been removed and updates the delivery status.

After the medicine is retrieved, the system sends the delivery information to the IoT cloud platform and stores the data using **SHA-256 hashing** to ensure data integrity. If authentication fails or the medicine is not retrieved within a predefined time, the system triggers an alert through the buzzer and updates the monitoring dashboard.

This sequence of operations ensures secure, automated, and reliable medicine delivery while maintaining accurate monitoring records.

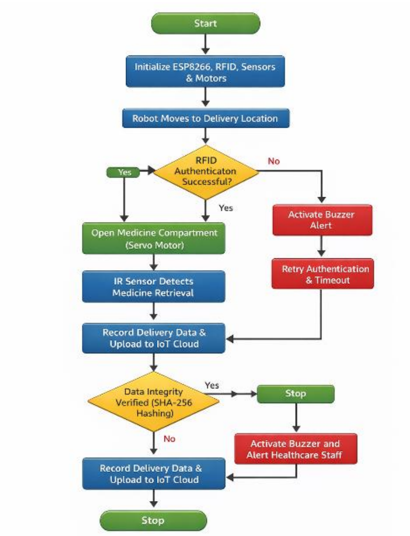

Fig. 5. Flowchart of the software operation

V. EXPERIMENTAL SETUP AND RESULTS

This section describes the experimental setup and performance evaluation of the proposed **Blockchain Enabled Secure Medical Delivery Robot Using IoT**. The objective of this experimental study is to validate the **accuracy, reliability, responsiveness, and operational efficiency** of the robotic medicine delivery system under simulated hospital conditions.

The hardware prototype consists of a **robotic delivery platform, RFID authentication module, IR sensor-based medicine detection mechanism, IoT communication module, and blockchain-inspired data security mechanism**. The system was tested under controlled laboratory conditions to evaluate parameters such as **robot navigation accuracy, RFID authentication success rate, medicine detection accuracy, IoT communication reliability, and system response time**.

The experimental results demonstrate that the proposed system provides **secure medicine delivery, reliable authentication, real-time monitoring, and stable system operation**, making it suitable for smart healthcare environments.

A. Experimental Setp

The hardware prototype of the proposed **medical delivery robot** was developed to simulate a real hospital medicine distribution scenario. The robot platform was built using **four DC motors connected to an L298N motor driver module**, which enables controlled movement of the robot along a predefined path.

The **ESP8266 NodeMCU microcontroller** serves as the central processing unit that manages robot navigation, RFID authentication, sensor monitoring, and IoT communication. An **RFID reader module** was installed on the robot to verify authorized users before allowing access to the medicine compartment.

A **servo motor mechanism** was integrated to control the opening and closing of the medicine compartment. Once the authorized RFID tag is detected, the servo motor rotates to open the compartment door. An **infrared (IR) sensor** is placed inside the compartment to detect whether the medicine has been removed.

The entire system is powered using a **regulated DC power supply**, which ensures stable voltage for all electronic components. The NodeMCU uses its **built-in Wi-Fi module** to transmit system data to the **ThingSpeak IoT cloud platform**, enabling real-time monitoring through a web dashboard.

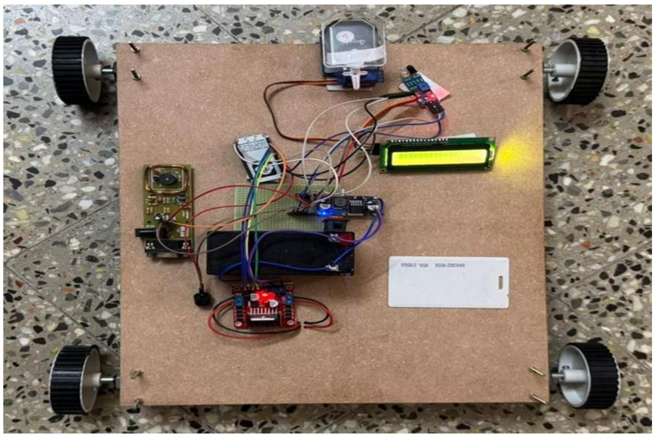

Fig. 6. Hardware prototype of the Blockchain Enabled Medical Delivery Robot

Fig. 7. IoT Sensor & NodeMCU Circuit Module

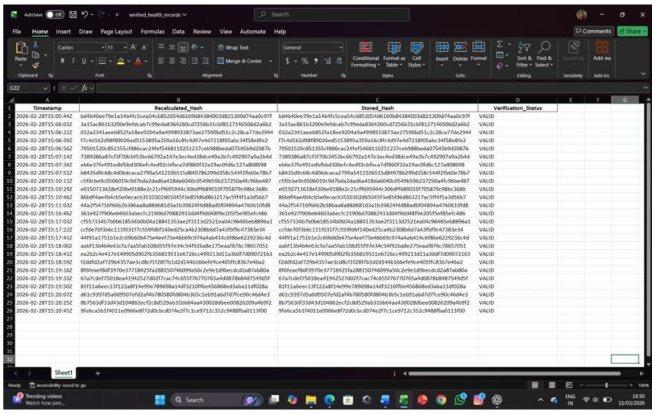

Fig. 8. Medical delivery verification results

B. Test Methodology

The system was tested under multiple operating scenarios to evaluate the reliability of robot navigation, authentication mechanisms, and IoT communication.

Robot navigation tests were conducted by allowing the robot to move along a predefined path to the medicine delivery location. The robot's ability to reach the destination without deviation was recorded across multiple trials.

RFID authentication tests were performed by scanning both **authorized and unauthorized RFID tags**. The system verified valid tags and rejected invalid authentication attempts to ensure secure access control.

Medicine retrieval detection was evaluated using the IR sensor by placing and removing objects inside the medicine compartment. The system response time for detecting medicine retrieval was recorded.

IoT communication tests were conducted by monitoring the transmission of delivery data and system status updates to the **ThingSpeak cloud platform**. Network interruptions were introduced to evaluate system stability and automatic reconnection capability.

C. Performance Metrics

The performance of the proposed system was evaluated using several quantitative metrics, including:

- Robot navigation accuracy
- RFID authentication success rate
- Medicine detection accuracy
- IoT communication reliability
- Data security verification
- System response time

Robot navigation accuracy was calculated by measuring the number of successful deliveries to the designated location. RFID authentication success rate was measured by comparing valid tag recognition with total authentication attempts.

IoT communication performance was evaluated by measuring the delay between sensor data acquisition and cloud dashboard updates. The SHA-256 hashing mechanism was tested to verify the integrity of stored delivery records.

D. System Performance Evaluation

The overall system performance was evaluated by analyzing the coordinated operation of robot navigation, authentication mechanisms, IoT communication, and data security features.

The robotic platform demonstrated **stable and repeatable navigation performance** across multiple test cycles. The robot successfully reached the delivery location in most trials without deviation from the predefined path.

The **RFID authentication system** reliably verified authorized tags and denied access for unauthorized users. The servo motor mechanism responded immediately after successful authentication, enabling smooth opening and closing of the medicine compartment.

The **IR sensor detection system** accurately identified whether the medicine had been retrieved from the compartment. The detection results were transmitted to the IoT platform in real time.

IoT communication through the NodeMCU's Wi-Fi module remained stable throughout extended runtime testing. Data updates appeared on the monitoring dashboard with minimal latency.

E. Monitoring Accuracy Evaluation

The monitoring accuracy of the system was evaluated by comparing sensor readings and delivery data with actual system events. The IR sensor demonstrated consistent performance in detecting medicine retrieval without false triggers.

RFID authentication accuracy remained high during multiple test trials. The IoT platform successfully recorded delivery events and system status updates without data loss.

The SHA-256 hashing mechanism ensured that stored data remained secure and tamper-resistant. Any modification of stored delivery records resulted in a change in the hash value, confirming the effectiveness of the blockchain-inspired security mechanism.

Parameter	**Key Components**	**Detection Accuracy (%)**	**Respons e Time (Sec)**
Robot Navigation	DC Motors + L298N	95.8	2.3
RFID Authenticati on	RFID Module	98.2	1.5
Medicine Detection	IR Sensor	96.4	1.2
Servo Compartmen t Control	Servo Motor	99.1	1.4
IoT Data Transmissio n	ESP8266 Wi-Fi	-	<1
Data Security Verification	SHA-256 Hashing	100	<1
Overall System Performance	Integrated System	95.2	<3

VI. DISCUSSION

The experimental evaluation of the proposed **Blockchain Enabled Secure Medical Delivery Robot Using IoT** demonstrates that the developed robotic system operates reliably under simulated hospital conditions. The integration of **robot navigation, RFID-based authentication, IoT communication, and blockchain-inspired data security** within a single embedded platform provides an efficient solution for automated medicine delivery. During experimental testing, the system showed stable performance without unexpected resets, communication failures, or inconsistencies between hardware and software modules, confirming strong system integration.

The robotic navigation system driven by **DC motors and the L298N motor driver** exhibited consistent movement along the predefined delivery path. Multiple test cycles confirmed that the robot could reach the intended delivery location with minimal deviation. The integration of controlled motor movement ensured smooth operation and reliable navigation suitable for hospital environments.

The **RFID authentication mechanism** demonstrated high accuracy in identifying authorized users. The system successfully verified valid RFID tags and rejected unauthorized attempts, ensuring secure access to the medicine compartment. This authentication process plays an important role in preventing unauthorized medicine retrieval and maintaining patient safety.

The **medicine retrieval detection mechanism using the IR sensor** performed reliably during testing. The sensor accurately detected the presence or removal of medicine from the compartment and updated the delivery status accordingly. This feature ensures that healthcare staff can confirm whether the medication has been successfully collected by the authorized person.

IoT communication using the **ESP8266 NodeMCU Wi-Fi module** remained stable throughout extended system operation. The web-based monitoring dashboard displayed delivery status and health monitoring parameters in real time with minimal latency. Even during temporary network interruptions, the system maintained local operation and automatically reconnected to the IoT platform without losing data synchronization.

The **blockchain-inspired SHA-256 hashing mechanism** effectively secured the stored delivery records and monitoring data. The hashing process ensured that any unauthorized modification of stored information could be detected, thereby improving the reliability and trustworthiness of the healthcare data management system.

Overall, the integration of robotic automation, secure authentication, IoT communication, and cryptographic data protection significantly improves hospital operational efficiency. The proposed system reduces the manual workload of healthcare staff, ensures secure medicine delivery, and enhances patient safety. These results indicate that the developed system can serve as a **cost-effective and scalable solution for smart hospital automation and healthcare logistics management**.

VII. CONCLUSION

This paper presented the design, implementation, and experimental evaluation of a **Blockchain Enabled Secure Medical Delivery Robot Using IoT**, developed to automate medicine delivery in hospital environments while ensuring secure access and reliable data monitoring. The system integrates **robotic navigation, RFID-based authentication, IoT communication, and blockchain-inspired data security** into a single intelligent healthcare automation platform. The primary objective of reducing manual workload for healthcare professionals while improving patient safety and operational efficiency has been successfully achieved.

The experimental results demonstrated that the robotic delivery system can reliably navigate along a predefined path and reach the intended delivery location with high accuracy. The **RFID authentication mechanism** effectively verified authorized users and prevented unauthorized access to the medicine compartment. The **IR sensor-based medicine retrieval detection system** confirmed whether the medication had been successfully collected, enabling accurate tracking of delivery events. The system also transmitted delivery status and health monitoring data to the IoT cloud platform, allowing healthcare professionals to monitor the system remotely through a web-based dashboard.

The integration of a **blockchain-inspired SHA-256 hashing mechanism** enhanced data integrity by ensuring that stored delivery records and monitoring data could not be modified without detection. This security mechanism improves the reliability and trustworthiness of healthcare data management systems. The overall system demonstrated stable operation, reliable communication, and efficient resource utilization during extended runtime testing.

From a technological perspective, the use of the **ESP8266 NodeMCU microcontroller** provides a cost-effective and energy-efficient platform with built-in Wi-Fi connectivity suitable for IoT-based healthcare applications. The modular design of the system allows easy integration of additional features such as obstacle detection, autonomous navigation, or advanced authentication mechanisms. This flexibility makes the system adaptable for different hospital environments and healthcare automation requirements.

In conclusion, the proposed system provides a **secure, automated, and contactless medicine delivery solution** for modern smart hospitals. By combining robotics, IoT monitoring, authentication mechanisms, and cryptographic data protection, the system enhances hospital efficiency, improves patient safety, and supports the development of intelligent healthcare infrastructures. Furthermore, the system contributes to reducing operational delays, improving medicine distribution accuracy, and enabling better management of hospital resources. With further improvements such as autonomous navigation, advanced sensors, and full blockchain integration, the proposed system has strong potential to become an important component of **next-generation smart healthcare systems** and large-scale hospital automation platforms.

ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to the **Department of Electronics and Communication Engineering** and the management of **Sri Shakthi Institute of Engineering and Technology, Coimbatore**, for providing the necessary infrastructure, laboratory facilities, and technical resources required to successfully carry out this research work. The authors also extend their heartfelt thanks to their **project guide and faculty members** for their continuous guidance, encouragement, and valuable suggestions throughout the development and testing phases of the proposed system. Their technical insights and support greatly contributed to the successful completion of this project. The authors are also grateful to the **technical staff and peers** for their assistance and cooperation during experimentation and system implementation.

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